Original Document

様邸

動的耐震診断システムによる耐震診断報告書

平成１６年３月

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Translation

Dynamic Seismic Diagnosis Report

Prepared Using the
Dynamic Seismic Diagnosis System

Date

March, 2004

Prepared by

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Readability Status

Readable (confirmed)

- Title is clearly readable.
- Date is clearly readable.
- Company name is clearly readable.
- Address, telephone number, and fax number are clearly readable.

Partially Difficult to Read

- None

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- None

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はじめに　～この報告書のみかた

加速度と震度階級

地震の揺れの強さを示すのに一般に使用されているものとして、気象庁が発表している「震度階級」があります。

しかし、これは診断結果として表示するには大まかにすぎること、また、約４００ｇａｌ以上のすべての地震が震度７と表示されることから、この報告書では、地震の揺れの強さ（加速度）を示すものとして、「ｇａｌ（ｃｍ／ｓ^2）」を用いています。

以下に、加速度と震度階級との関係を表にしています。これを参考にしながら報告書をご覧ください。

また、地震の規模を表すのに、「マグニチュード」という語が使われます。「マグニチュード」は地震そのもののエネルギーの大きさを表すもので、「加速度」や「震度階級」は調査地での揺れの大きさを表すものです。

ちなみに、兵庫県南部地震（阪神大震災）の地震の規模はマグニチュード７．２、震源から約２５ｋｍ離れた神戸海洋気象台では８１８ｇａｌの揺れを記録しています。

加速度と震度階級との関係

加速度(gal)	震度階級	
～0.8	0	人は揺れを感じない
0.8～2.5	1	屋内にいる人の一部が、わずかな揺れを感じる
2.5～8.0	2	屋内にいる人の多くが、揺れを感じる。眠っている人の一部が目を覚ます。
8.0～25	3	屋内にいる人のほとんどが揺れを感じる。恐怖感を覚える人もいる。
25～80	4	かなりの恐怖感があり、一部の人は身の安全を守ろうとする。眠っている人のほとんどが目を覚ます。
80～250	5弱	多くの人が身の安全を図ろうとする。一部の人は行動に支障を感じる。
	5強	非常な恐怖感を感じる。多くの人が行動に支障を生じる。
250～400	6弱	立っていることが困難になる。
	6強	立っていることができず、はわないと動くことが出来ない。
400～	7	揺れにほんろうされ、自分の意志で行動できない。

※加速度と震度階級は、地震の継続時間等諸条件により必ずしも一致しない場合があります。

Translation

Introduction – How to Read This Report

Acceleration and Seismic Intensity Scale

One commonly used indicator of the strength of earthquake shaking is the "seismic intensity scale" announced by the Japan Meteorological Agency.

However, this scale is too approximate to be used as a diagnostic result, and since all earthquakes of 400 gal or more are indicated as seismic intensity 7, this report uses "gal (cm/s^2)", which indicates the strength of earthquake shaking (acceleration).

The following table shows the relationship between acceleration and the seismic intensity scale. Please refer to this table while reading the report.

The term "magnitude" is also used to indicate the scale of an earthquake. "Magnitude" expresses the size of the earthquake energy itself, while "acceleration" and "seismic intensity scale" express the shaking level at the survey site.

By the way, the magnitude of the 1995 Hyogo-ken Nanbu Earthquake (the Great Hanshin Earthquake) was 7.2, and a shaking of 818 gal was recorded at the Kobe Marine Observatory, located about 25 km from the epicenter.

Relationship between Acceleration and Seismic Intensity Scale

Acceleration (gal)	Seismic Intensity Scale	
~0.8	0	People do not feel any shaking.
0.8~2.5	1	Some people indoors feel a slight shaking.
2.5~8.0	2	Many people indoors feel the shaking. Some people who are sleeping wake up.
8.0~25	3	Most people indoors feel the shaking. Some people feel fear.
25~80	4	Considerable fear is felt, and some people try to protect themselves. Most people who are sleeping wake up.
80~250	5-	Many people try to ensure their safety. Some people experience difficulty in their actions.
	5+	People feel extreme fear. Many people have difficulty in their actions.
250~400	6-	It becomes difficult to remain standing.
	6+	Unable to stand and must crawl to move.
400~	7	Overwhelmed by the shaking and unable to act according to one's own will.

* Acceleration and seismic intensity scale may not always correspond exactly, depending on factors such as the duration of the earthquake.

Readability Status

Readable (confirmed)

- All text is clearly readable.
- Table contents are clearly readable.
- Footnote is clearly readable.

Partially Difficult to Read

- None

Not Readable / Unclear

- None

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診断日:2004年3月16日

計　測　内　容

1. X方向・Y方向起振試験　・・・　P2～P7
2. X方向起振試験　・・・　P8
3. 常時微動計測　・・・　P9～P10
4. 表面波探査法による地盤固有周期調査　・・・　P1

計　測　結　果

建物の耐震性能

X方向	共振周波数	損傷限界	安全限界	減衰定数
南側	6.2Hz	418.8gal	651.8gal	0.0535
中央部	6.2Hz	522.9gal	813.9gal	0.0676
北側	6.2Hz	677.9gal	1055gal	0.0865

地盤の分析

固有周期	0.26秒（3.9Hz）
地震動の増幅率	1.78倍
地盤種別	第２種地盤

Translation

Mr. XXXXX

Inspection Date: March 16, 2004

Measurement Contents

1. Excitation Test in X and Y Directions ・・・ Pages P2–P7
2. Excitation Test in X Direction ・・・ Page P8
3. Microtremor Measurement ・・・ Pages P9–P10
4. Ground Natural Period Investigation by Surface Wave Exploration Method ・・・ Page P1

Measurement Results

Seismic Performance of Building

X Direction	Resonant Frequency	Damage Limit	Safety Limit	Damping Ratio
South Side	6.2 Hz	418.8 gal	651.8 gal	0.0535
Center	6.2 Hz	522.9 gal	813.9 gal	0.0676
North Side	6.2 Hz	677.9 gal	1055 gal	0.0865

Ground Analysis

Natural Period	0.26 sec (3.9 Hz)
Amplification Factor of Seismic Motion	1.78
Ground Classification	Type II Ground

Readability Status

Readable (confirmed)

- Inspection date is clearly readable.
- Measurement contents are clearly readable.
- All numerical values in the table are clearly readable.
- Ground analysis values are clearly readable.

Partially Difficult to Read

- None

Not Readable / Unclear

- None

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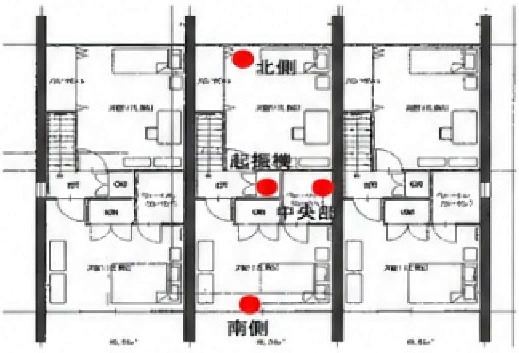
＜ 計測１ X方向の耐震性計測 ＞

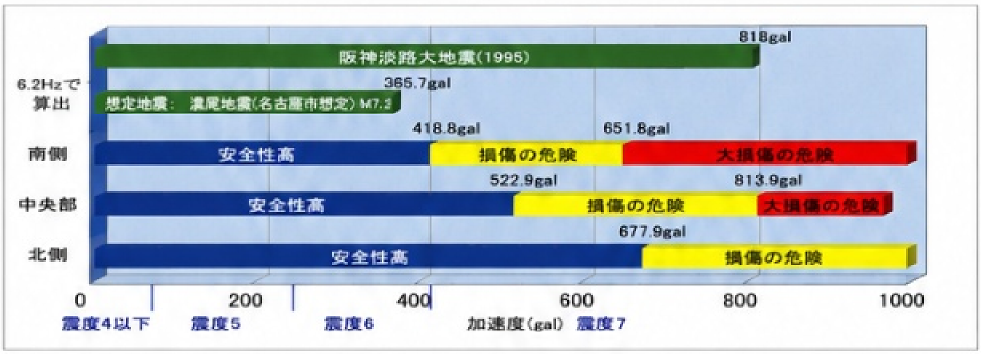


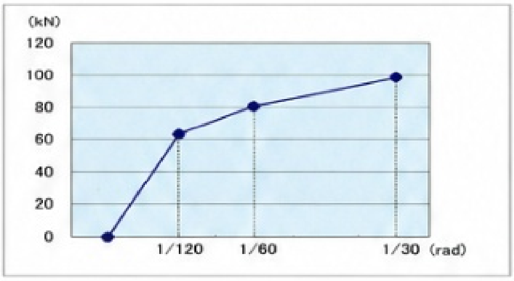
X方向に起振して、加速度検出器を南・中央・北の3点に配置して計測した。

設置高さはすべて310cmである。

分析結果は以下の通りである。
1階部に壁の多い北側が、南側に比べやや揺れにくいという特徴がみられる。

分 析 結 果 ＜ X 方 向 （ 短 辺 ・ 東 西 方 向 ） ＞



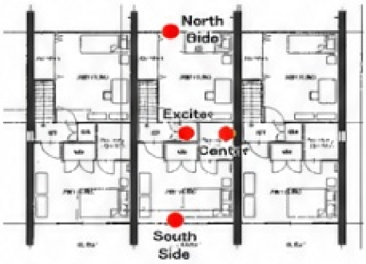


上の分析結果は、当社が平成15年に行った、木造住宅の荷重ー変形試験結果をもとにした推測値である。

起振試験により得た結果を、1/120radの変位までを弾性範囲と仮定し、「安全性の高い範囲」として計算、1/30radの変位までを左図に基づき計算し、「損傷の危険」、1/30radより大きな変位を来たすと推測される範囲を、「大損傷の危険」と表示している。

Translation

< Measurement 1 Seismic Performance Measurement in X Direction >

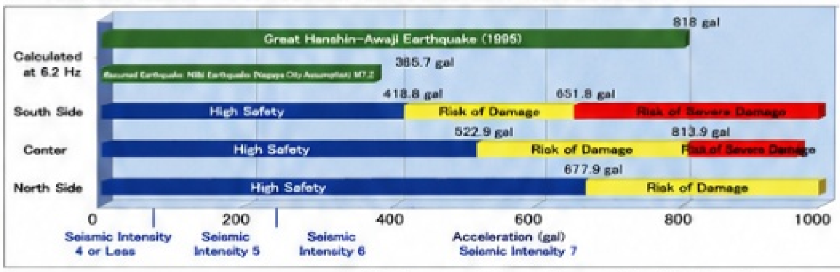


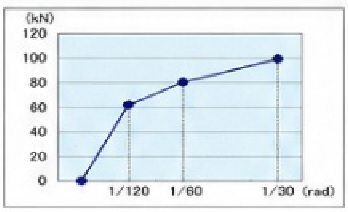
Excitation was applied in the X direction, and accelerometers were placed at three points: south, center, and north, and measurements were taken.

The installation height is 310 cm for all points.

The analysis results are as follows.
It is observed that the north side, which has many walls on the first floor, tends to shake slightly less compared to the south side.

Analysis Results < X Direction (Short Side: East-West Direction) >





The above analysis results are estimated values based on the load-deformation test results of a wooden house conducted by our company in 2003 (Heisei 15).

The results obtained from the excitation test are calculated assuming the elastic range up to a displacement of 1/120 rad as the "high safety range." Up to a displacement of 1/30 rad, the values are calculated based on the graph on the left and indicated as "risk of damage." The range where larger displacements than 1/30 rad are expected is indicated as "risk of severe damage."

Readability Status

Readable (confirmed)

- All text is clearly readable.
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- Numerical values and units are clearly readable.
- Graphs and axis labels are clearly readable.

Partially Difficult to Read

- None

Not Readable / Unclear

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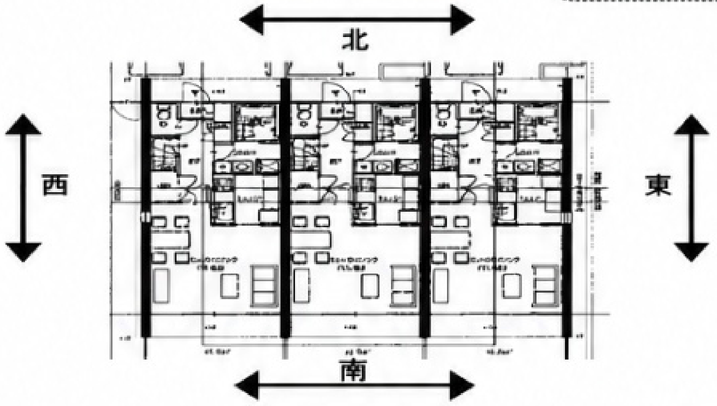
建物各面の耐震性能

調査物件において、何galの地震で建物1階の最大変位が1/120radを超えるか（1階天井高さ2.4mの場合2cm）を建物の東西南北各面において示した図です。

濃青	800gal以上	阪神淡路大地震クラス
水色	600gal～800gal	震度7クラス
黄	400gal～600gal	
赤	400gal未満	震度6以下クラス

677.9gal



418.8gal

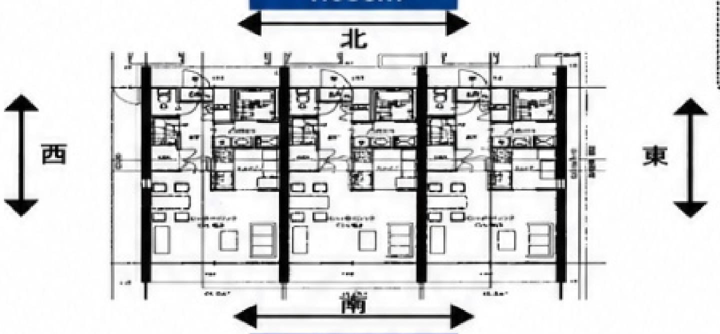
想定地震における1階壁の最大ひずみ量

各地域における想定地震が仮に想定どおりに起きた場合の、建物1階部の最大変位量を表した図です。
1階天井高さ2.4mの場合、2cmを超えると損傷の危険が、4cmを超えると大きな損傷の危険が発生すると考えられます。

想定地震：濃尾地震（名古屋市想定
地震の規模 マグニチュードM7.2
震源からの距離30km

色	
濃青	安全性高
黄	損傷の危険
赤	大損傷の危険

1.08cm



1.75cm

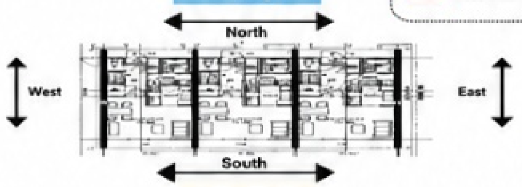
Translation

Seismic Performance of All Sides of the Building

This figure shows the seismic intensity level (in gal) at which the maximum displacement on the first floor exceeds 1/120 rad (2 cm when the first floor ceiling height is 2.4 m) for each side of the building (east, west, south, and north).

Dark blue	800 gal or more	Great Hanshin-Awaji Earthquake class
Light blue	600 gal to 800 gal	JMA Seismic Intensity 7 class
Yellow	400 gal to 600 gal	
Red	Less than 400 gal	JMA Seismic Intensity 6 or lower class

677.9 gal



418.8 gal

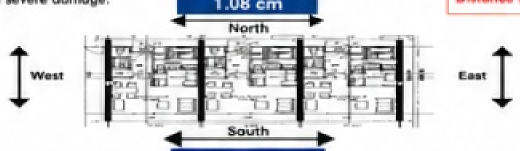
Maximum Strain of First-Floor Walls under Assumed Earthquake

This figure shows the maximum displacement on the first floor of the building when the assumed earthquake for each region occurs as anticipated.
When the first-floor ceiling height is 2.4 m, if the displacement exceeds 2 cm, there is a risk of damage, and if it exceeds 4 cm, there is a risk of severe damage.

Assumed Earthquake:
Nobi Earthquake (Nagoya City Assumption)
Magnitude: 7.2
Distance from Epicenter: 30 km

Dark blue	High Safety
Yellow	Risk of Damage
Red	Risk of Severe Damage

1.08 cm



1.75 cm

Readability Status

Readable (confirmed)

- All text is clearly readable.
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- Numerical values and units are clearly readable.
- Legend labels are clearly readable.

Partially Difficult to Read

- None

Not Readable / Unclear

- None

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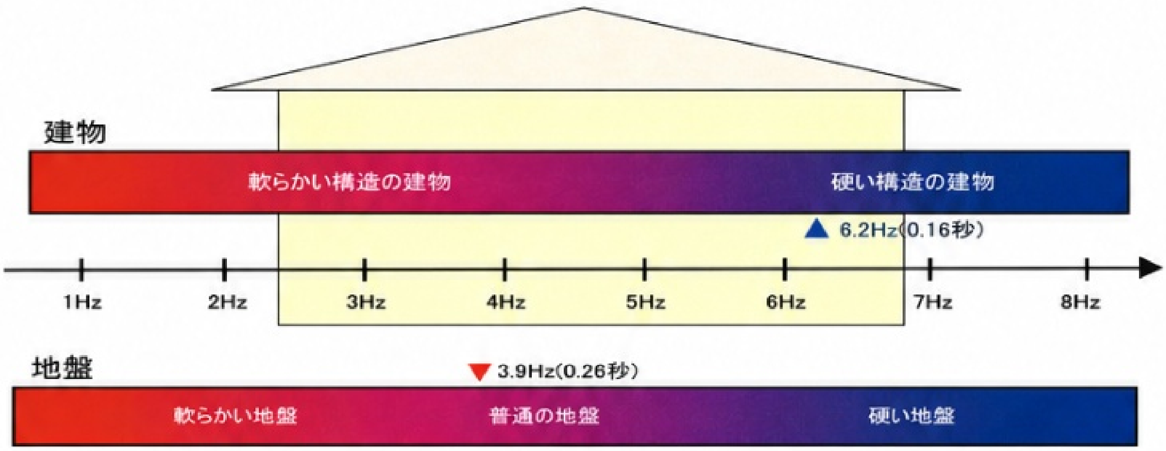
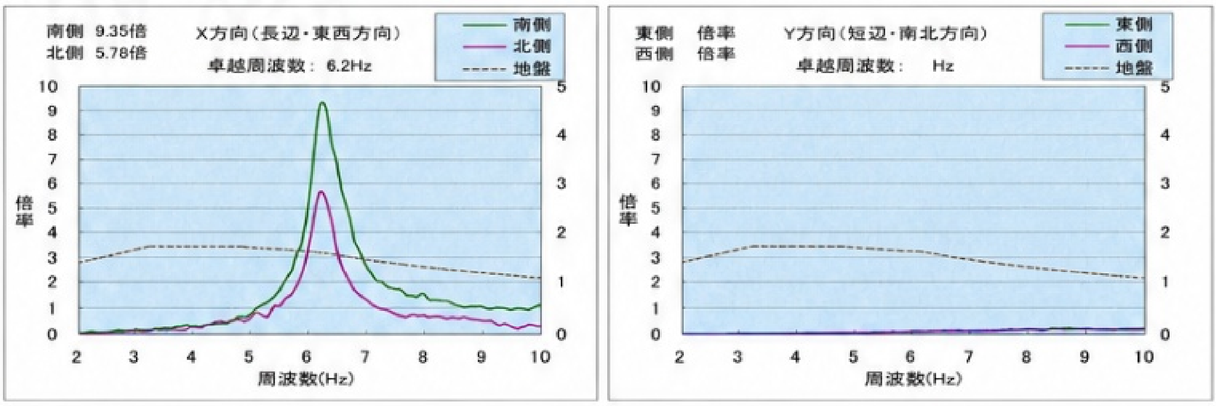
地盤と建物の卓越周波数比較

地盤と建物のもっとも揺れやすい周波数をあらわした図です。
地盤と建物のもっとも揺れやすい周波数が近いと、地震時に共振現象により建物が大きく揺れやすくなります。
共振のおそれがある場合、耐震改修の必要性がより高いと考えられます。

▼ 地盤の卓越周波数
▲ 建物X方向の卓越周波数
▲ 建物Y方向の卓越周波数



＜解析資料＞　増幅率グラフ



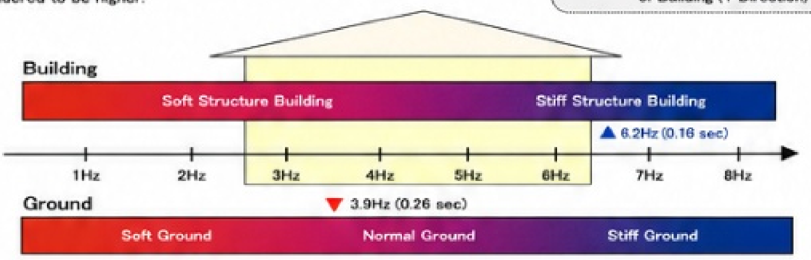
-4-

Translation

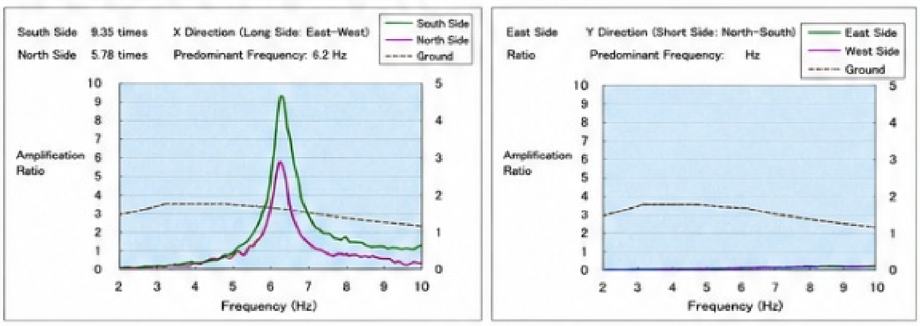
Comparison of Predominant Frequencies between Ground and Building

This figure shows the most likely-to-vibrate (predominant) frequencies of the ground and the building.
When the predominant frequencies of the ground and the building are close to each other, resonance may occur during an earthquake, causing the building to shake more severely.
If there is a risk of resonance, the need for seismic retrofitting is considered to be higher.

▼ Predominant Frequency of Ground
▲ Predominant Frequency of Building (X Direction)
▲ Numericant Frequency of Building (Y Direction)



< Analysis Information > Amplification Ratio Graphs



-4-

Readability Status

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- Graphs and axis labels are clearly readable.

Partially Difficult to Read

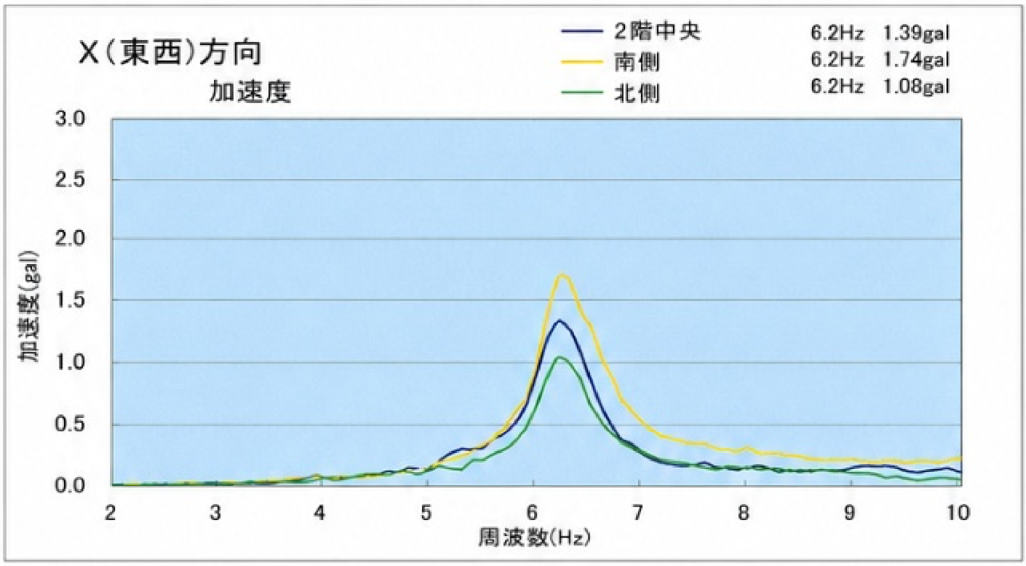
- None

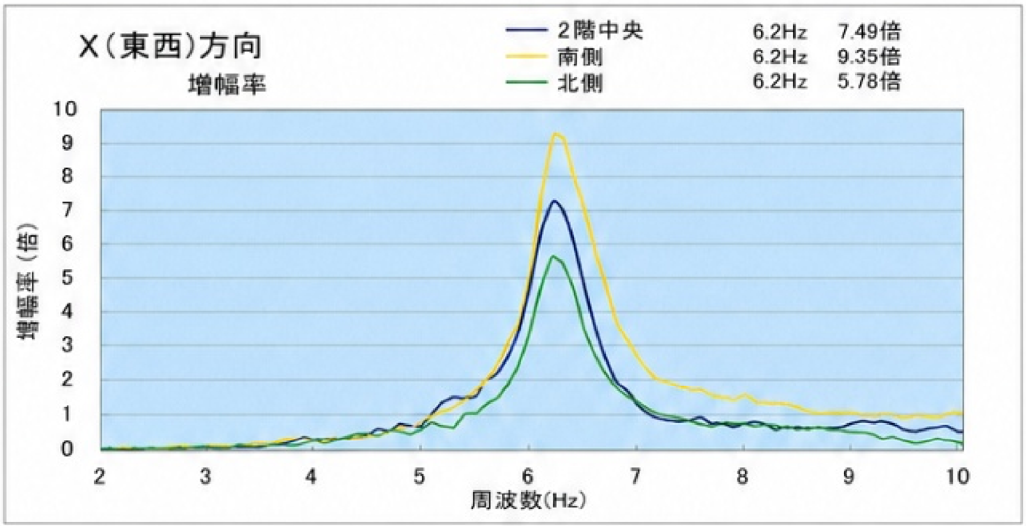
Not Readable / Unclear

- None

Original Document

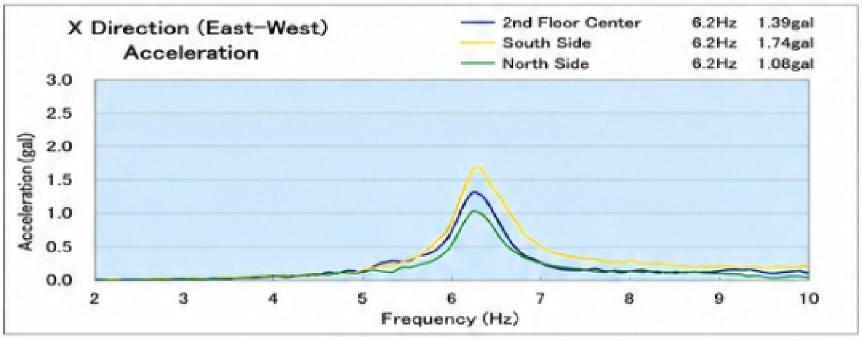
計測データ

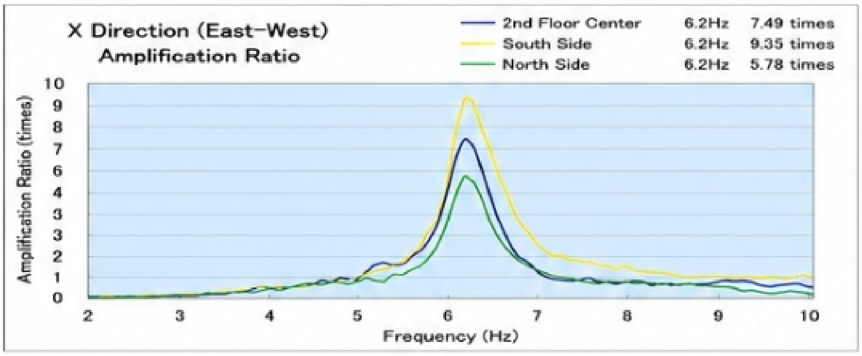




Translation

Measurement Data





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Partially Difficult to Read

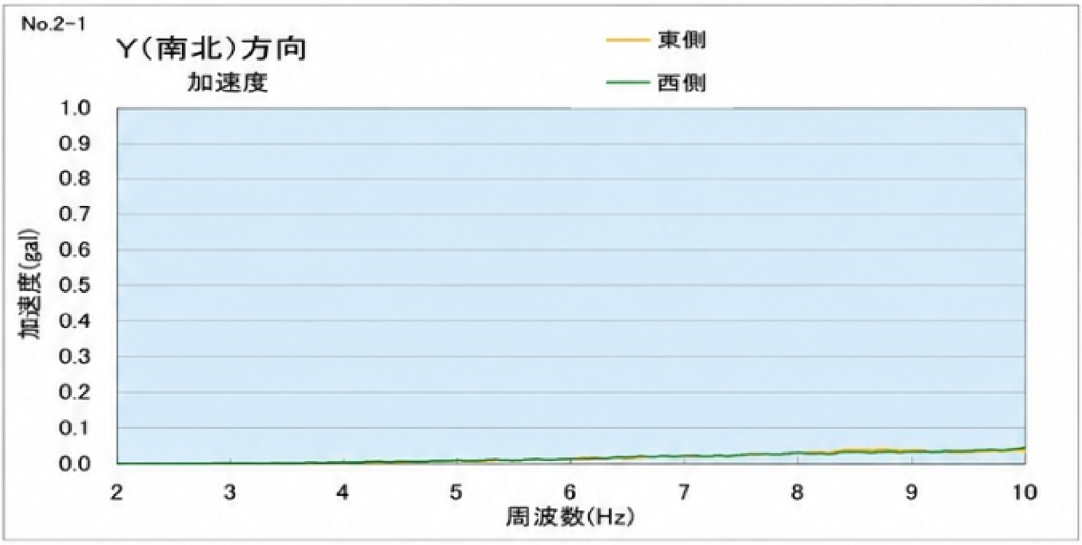
- None

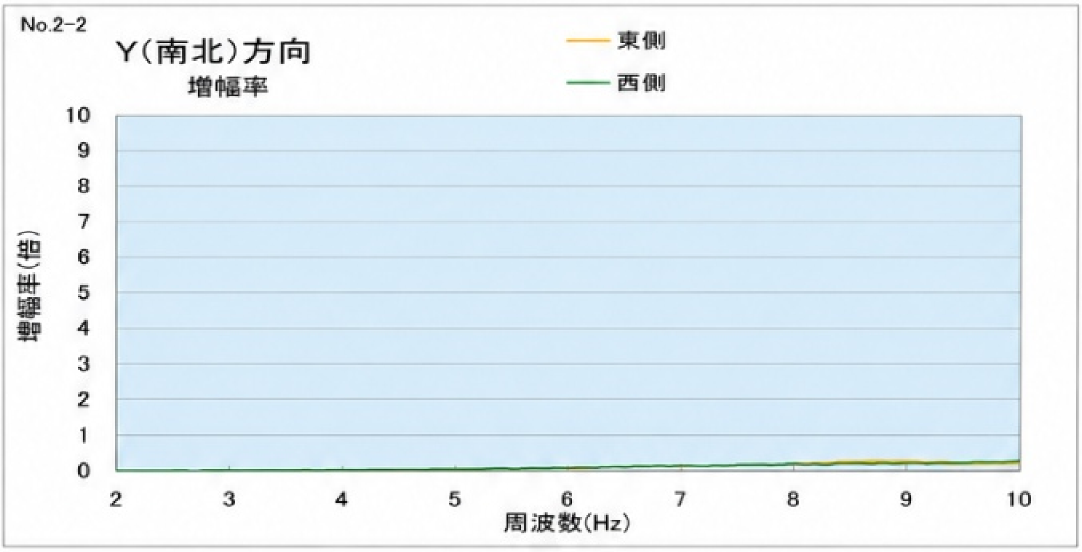
Not Readable / Unclear

- None

Original Document

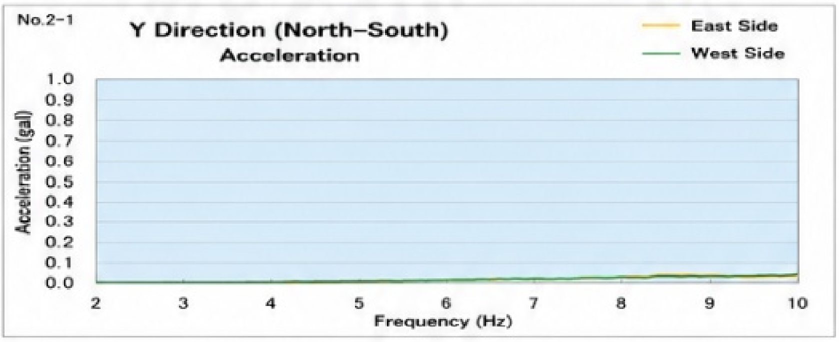
（Y方向については、2Hzから20Hzまでの間に、共振周波数がみられなかったため、分析の対象としていない）

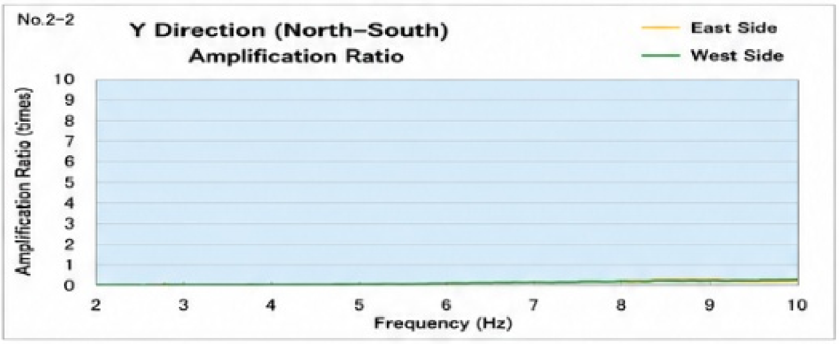




Translation

(Since no resonance frequency was observed in the Y direction between 2 Hz and 20 Hz, it was not included in the analysis.)





Readability Status

Readable (confirmed)

- All text is clearly readable.
- Diagrams and labels are clearly readable.
- Numerical values and units are clearly readable.
- Graphs and axis labels are clearly readable.

Partially Difficult to Read

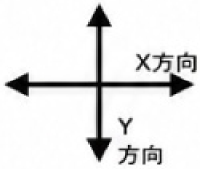
- None

Not Readable / Unclear

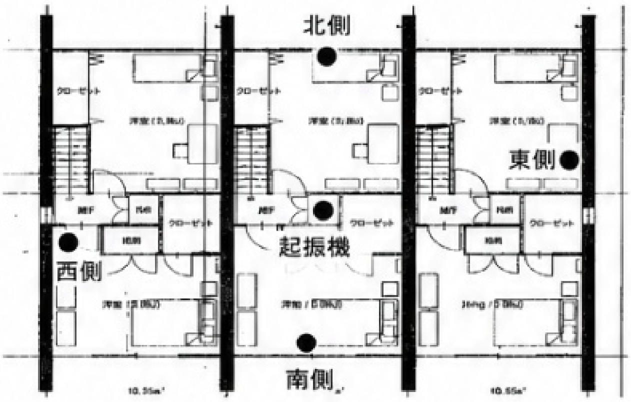
- None

Original Document

測　定　位　置



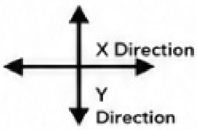
1 階平面図

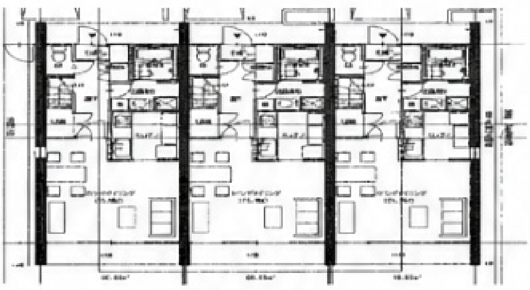


2 階平面図

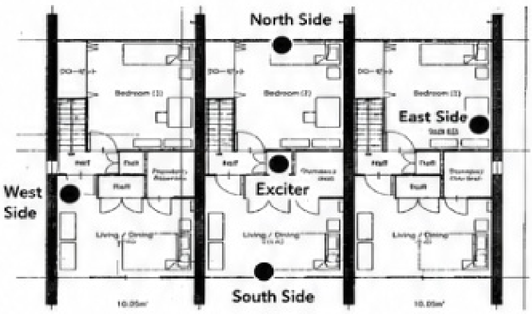
Translation

Measurement Positions





1st Floor Plan



2nd Floor Plan

Readability Status

◯ Readable (confirmed)

- All text is clearly readable.
- Diagrams and labels are clearly readable.
- Measurement positions are clearly indicated.
- Legend (X and Y directions) is clearly readable.

◯ Partially Difficult to Read

- None

◯ Not Readable / Unclear

- None

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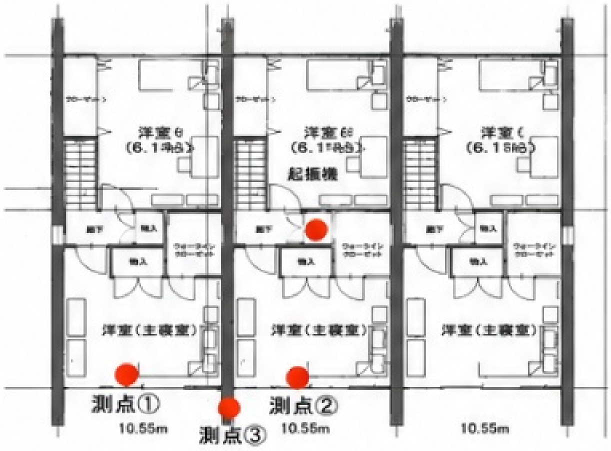
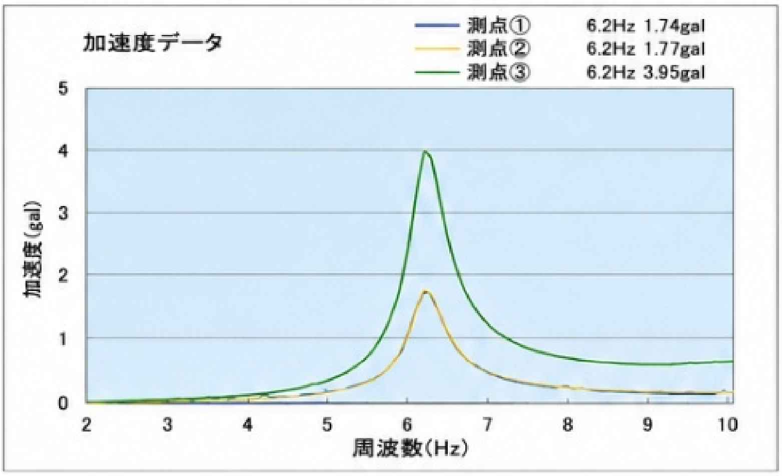
＜計測2 ～南側3箇所に検出器を設置して計測＞



X方向起振時に、起振機が建物全体に振動を伝えているかを確認するため、加速度検出器を左図の測点①～③位置に設置した。

設置高さは測点①および測点②が310cm、測点③が600cmである。

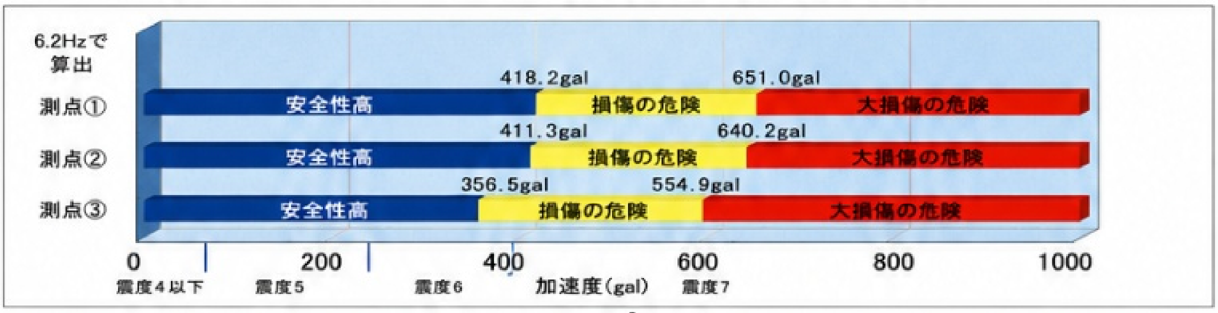


計測データは左の通りであり、測点①と測点②とはほとんど同じ特性を示しており、建物全体が揺れていることがわかる。

また、測点③の加速度値が大きいのは、設置高さが測点①②と異なるためと考えられる。

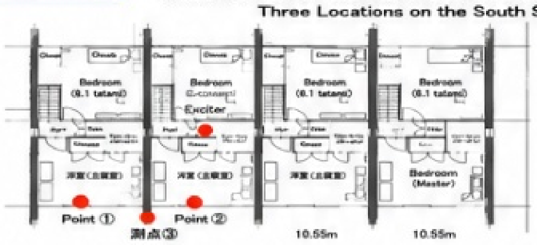
なお、設置高さを考慮に入れ、建物1階部の耐震性グラフで示すと下図のようになる。

分 析 結 果 ＜ X 方 向 3 点 で 計 測＞



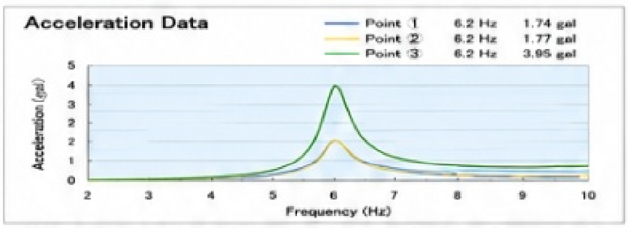
Translation

<Measurement 2 – Measurement with Detectors Installed at Three Locations on the South Side>



To confirm whether the exciter transmits vibration throughout the building during excitation in the X direction, acceleration detectors were installed at measurement points ① to ③ shown in the figure on the left.

The installation heights are 310 cm for points ① and ②, and 600 cm for point ③.

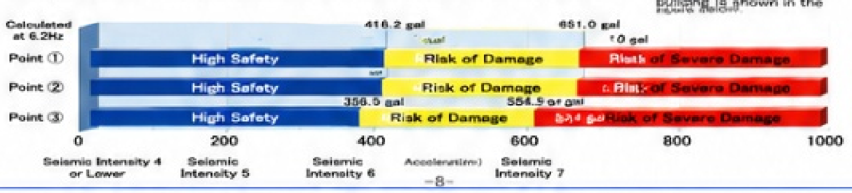


As shown in the data on the left, points ① and ② exhibit almost the same characteristics, indicating that the entire building is vibrating.

The higher acceleration value at point ③ is considered to be due to a different installation height from points ① and ②.

Considering the installation height, the seismic performance of the first floor of the building is shown in the figure below.

Analysis Results <Measurement at Three Points in X Direction>



Readability Status

Readable (confirmed)

- All text is clearly readable.
- Diagrams and labels are clearly readable.
- Numerical values and units are clearly readable.
- Legends and axis labels are clearly readable.

Partially Difficult to Read

- None

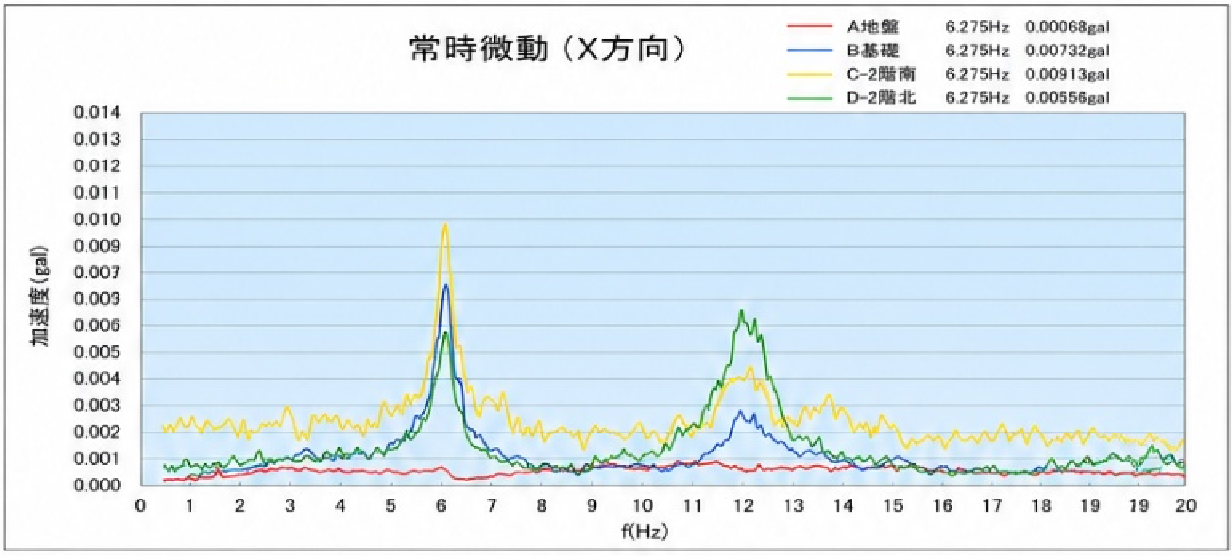
Not Readable / Unclear

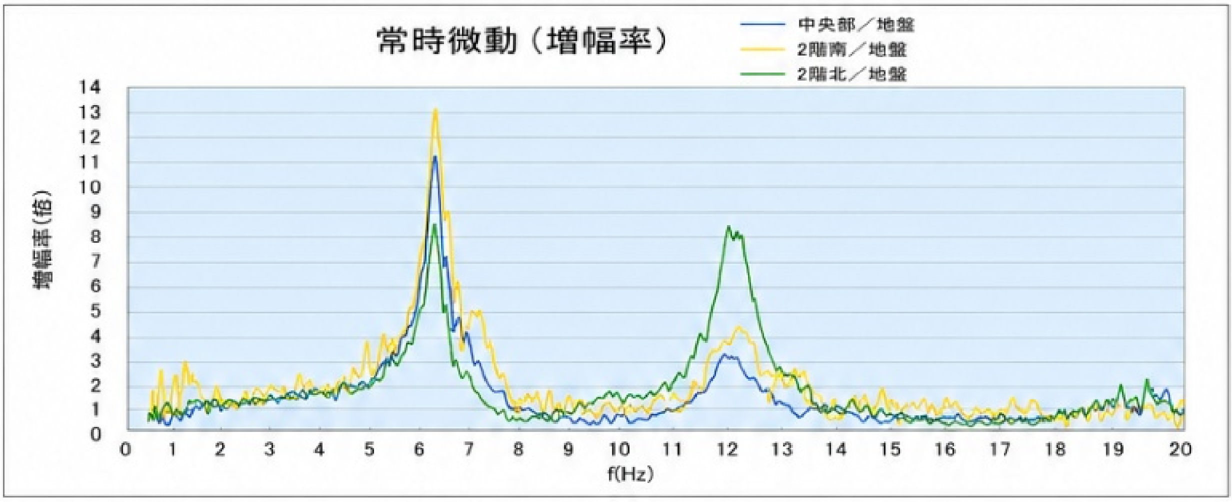
- None

Original Document

< 計測3　常時微動計測 >

起振試験結果の整合性確認のため、常時微動計測も併せて行った。
X方向においては、1階部に壁の多い北側が、南側に比べやや揺れにくいという特徴がみられる。

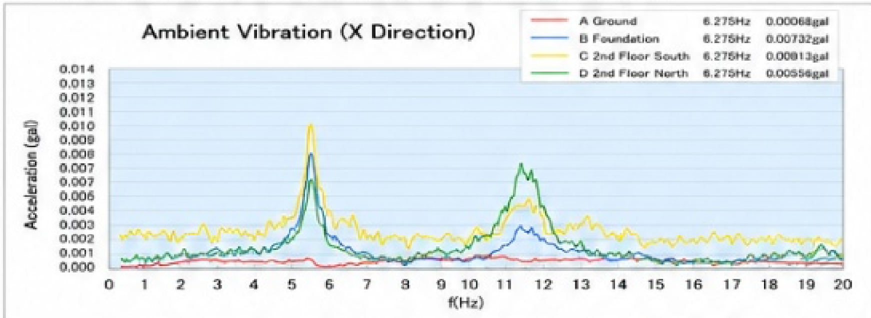


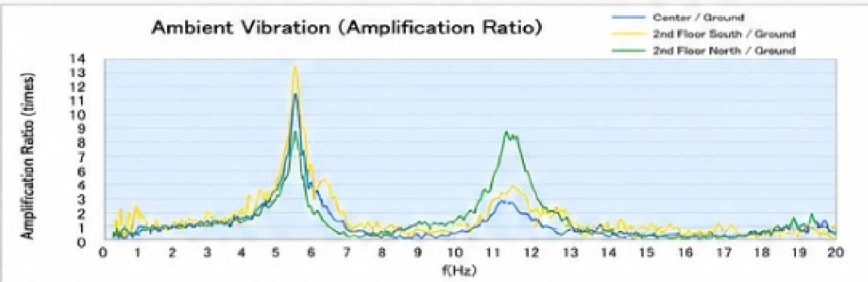


Translation

< Measurement 3: Ambient Vibration Measurement >

In order to verify the consistency of the excitation test results, ambient vibration measurements were also conducted.
In the X direction, it was observed that the north side, which has many walls on the first floor, tends to vibrate slightly less than the south side.





-9-

Readability Status

Readable (confirmed)

- All text is clearly readable.
- Diagrams and labels are clearly readable.
- Numerical values and units are clearly readable.
- Legends and axis labels are clearly readable.

Partially Difficult to Read

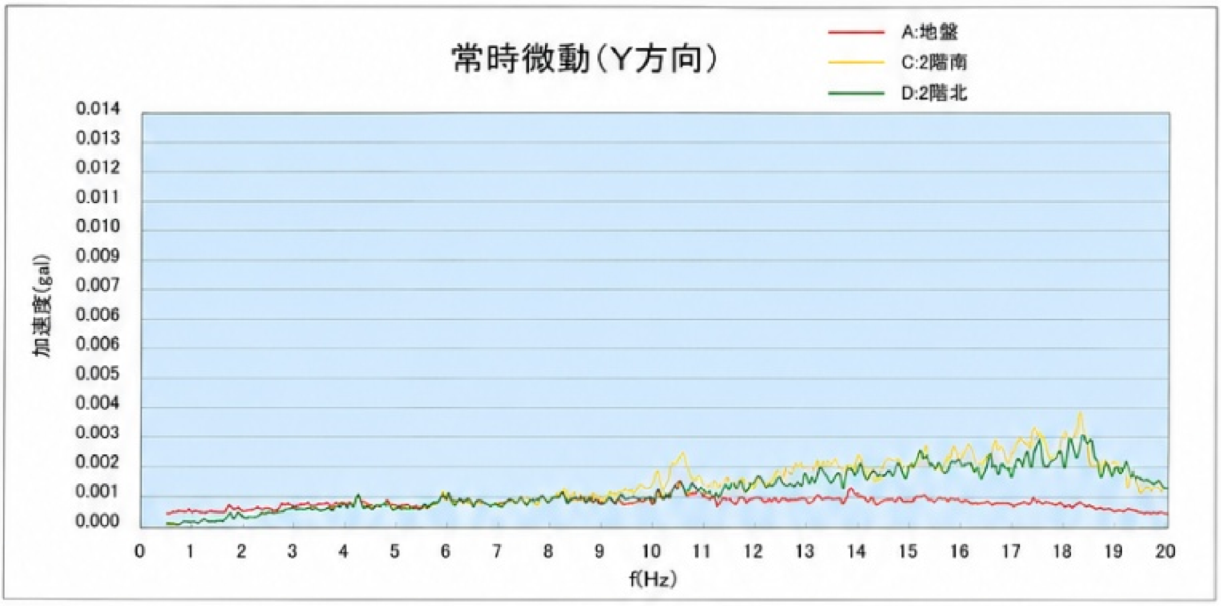
- None

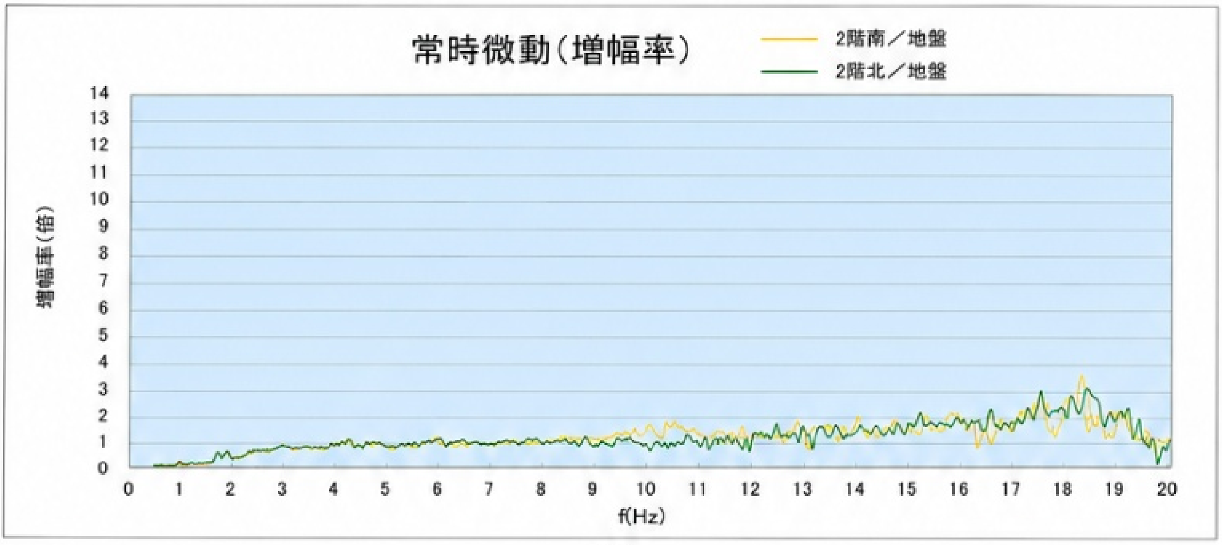
Not Readable / Unclear

- None

Original Document

Y方向においては、起振試験同様、2Hzから20Hzまでの間に、共振周波数を確認することができない。

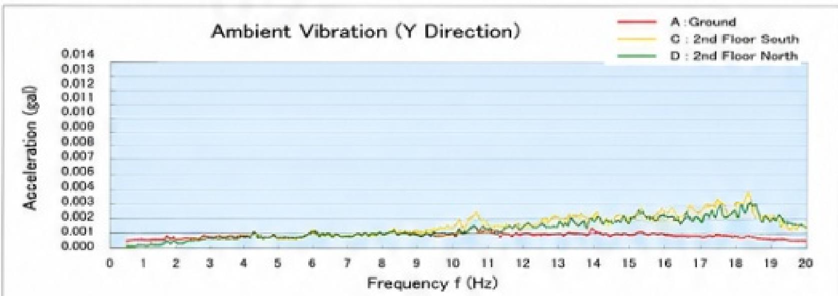


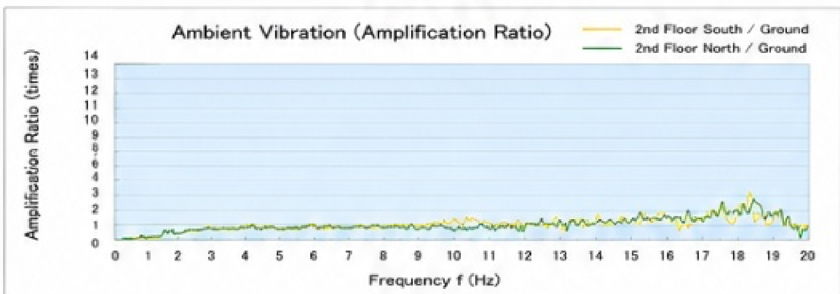


-10-

Translation

In the Y Direction,
as in the excitation test,
no resonant frequency
could be identified in the range
from 2 Hz to 20 Hz.





-10-

Readability Status

Readable (confirmed)

- All text is clearly readable.
- Graph titles are clearly readable.
- Legends and axis labels are clearly readable.
- Numerical values are clearly readable.

Partially Difficult to Read

- None

Not Readable / Unclear

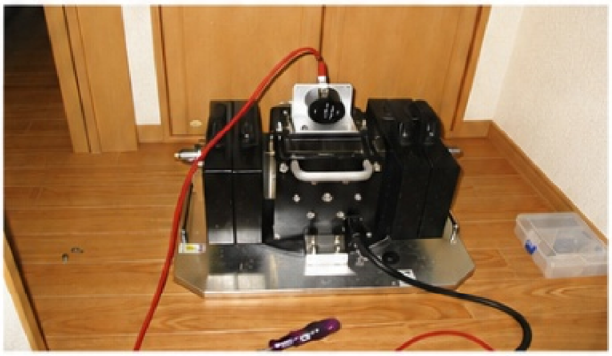
- None

Original Document

調　　査　　物　　件

測　　定　　機　　器

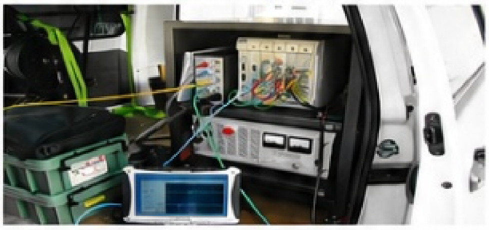
測　　定　　状　　況



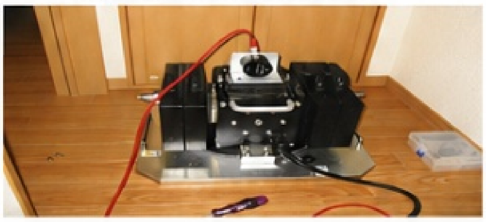
Translation

Subject Building

Measurement Equipment



Measurement Situation



-11-

Readability Status

Readable (confirmed)

- All text is clearly readable.
- Images are clear and legible.
- Layout is clear and well organized.
- No important information seems to be missing.

Partially Difficult to Read

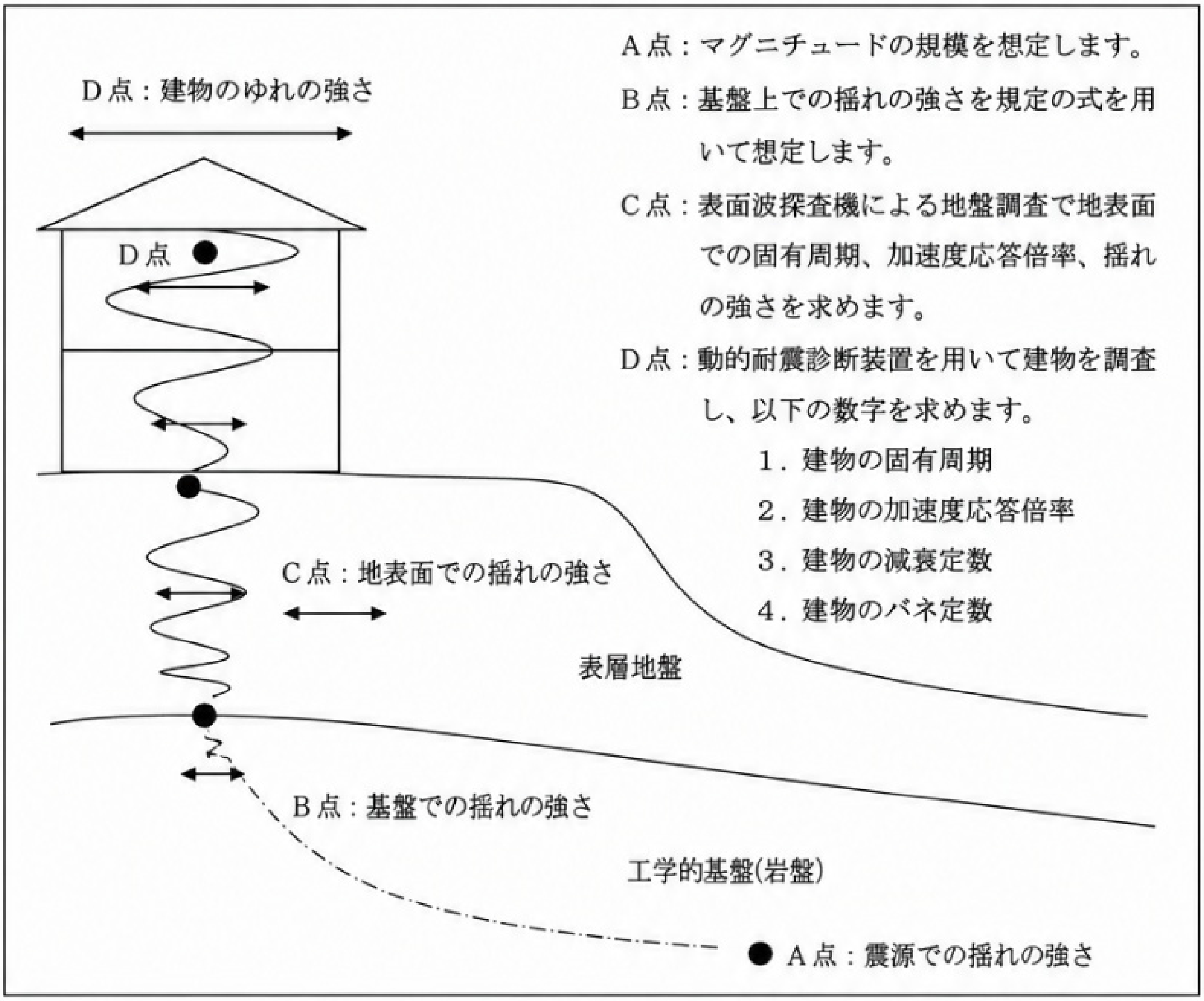
- None

Not Readable / Unclear

- None

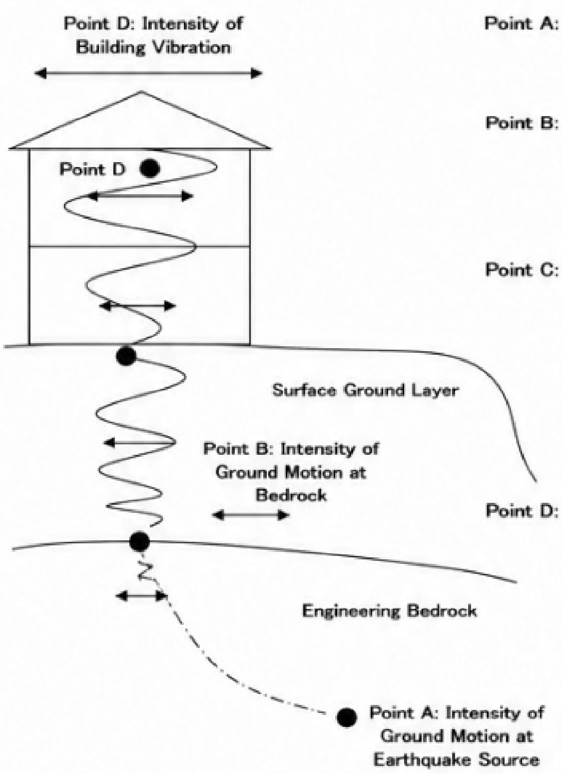
Original Document

＜動的耐震診断法の原理＞



Translation

Principle of the Dynamic Seismic Diagnosis Method



Point A: Assumed Magnitude of the Earthquake Source
The scale of the earthquake (magnitude) is assumed.

Point B: Intensity of Ground Motion at Bedrock
The intensity of shaking at the engineering bedrock is estimated using a prescribed formula.

Point C: Intensity of Ground Motion at the Ground Surface
A surface-wave exploration system is used to investigate the ground and determine:
- Natural period at the ground surface
- Acceleration response amplification ratio
- Intensity of ground motion

Point D: Intensity of Building Vibration
The building is investigated using the Dynamic Seismic Diagnosis System, and the following parameters are obtained:
1. Natural Period of the Building
2. Acceleration Response Amplification Ratio
3. Damping Ratio
4. Spring Constant of the Building

Readability Status

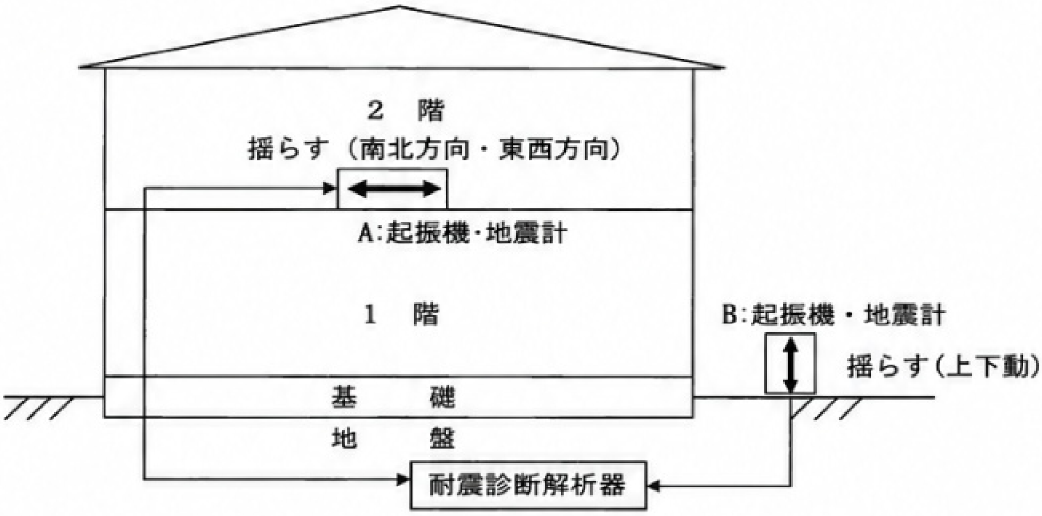
Readable (confirmed)
- Title is clearly readable.
- All explanatory text is clearly readable.
- Diagram labels are clearly readable.
- Ground layer names are clearly readable.
- No missing or unclear text was found.

Partially Difficult to Read
- None

Not Readable / Unclear
- None

Original Document

＜動的耐震診断法の模式図＞



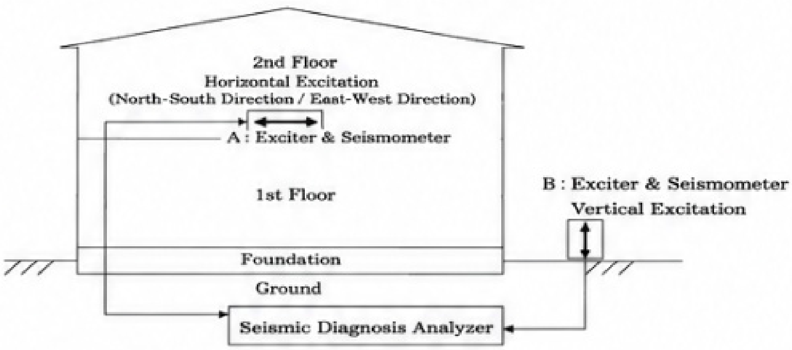
＊起振機Ａは建物を水平方向に振動させます。
地震計と組み合わせて建物の固有周期、地震時の加速度応答倍率（揺れの増幅度合い）、建物の減衰定数、建物のばね定数などを測定し、軽微な損傷あるいは損傷限界値における震度等を求めます。

＊起振機Ｂは地盤用で鉛直方向に振動させます。
地震計と組み合わせて地盤の固有周期、地震時の加速度応答倍率などを測定します。

＊耐震診断解析器は起振機の制御や地震波の取り込みを行い、建物と地盤の共振の可能性など様々な解析を行います。

Translation

Schematic Diagram of the Dynamic Seismic Diagnosis Method



* Exciter A vibrates the building horizontally.
  Combined with seismometers, it measures:
  - Natural period of the building
  - Acceleration response amplification ratio during earthquakes (degree of vibration amplification)
  - Damping ratio of the building
  - Spring constant of the building

  It also estimates the seismic intensity corresponding to slight damage or the damage limit state.

* Exciter B vibrates the ground vertically.
  Combined with seismometers, it measures:
  - Natural period of the ground
  - Acceleration response amplification ratio during earthquakes

* The Seismic Diagnosis Analyzer controls the exciters and acquires earthquake-wave data.
  It performs various analyses, including:
  - Possibility of resonance between the building and the ground
  - Dynamic characteristics of the structure
  - Seismic performance evaluation

Readability Status

Readable (confirmed)

- Title is clearly readable.
- Diagram labels are clearly readable.
- All explanatory text is clearly readable.
- No missing or unclear text was found.

Partially Difficult to Read

- None

Not Readable / Unclear

- None

Original Document

<動的耐震診断において用いる計算式>

1．想定条件で用いている地震の大きさに関しては以下の式を利用しています。

司・翠川・松岡は、地盤特性の影響を定量的に評価し、震源特性を震源深さも考慮して評価し、震源近傍での記録を含む最新の日本の強震記録の回帰分析により、震源近傍にも適用できる地震動最大振幅の予測式を提案している。

距離減衰式を求めるために用いた回帰モデルを以下に示す。

$$\log A = b_i - \log(X + c_i) - kX$$

但し、Aは最大振幅値、Xは断層面からの最短距離（km）、kは粘性減衰係数、b_i、c_iはi地震に対して求まる回帰係数である。

次に、最大加速度の距離減衰式を地盤の影響を考慮せずに求めた。c_iとb_iの結果は以下に示す。kは０．００３に固定している。

$$c_i = 0.000493 \quad 10^{0.608Mw}$$

$$b_i = 0.495M_w + A_D + 0.479$$

ここで、M_wはモーメントマグニチュード、A_Dは地震の震源深さが３０km以浅であれば０、３０km以上６０km未満ならば０．１８６、６０km以上では．０.５５４を与える。

（兵庫県南部地震を含む日本のデータに基づく最大地動加速度・速度の距離減衰式の予備的研究
司宏俊・翠川三郎・松岡昌志　１９９６）

2．表層地盤の固有周期及び増幅度

1）建設省告示第１４５７号によると表層地盤の一次卓越周期及び二次卓越周期は、それぞれ次に掲げる式によって計算する。

（1）$T_1 = 4 \times (\Sigma H_i)^2 / (\Sigma(((G_i/\rho_i) \times H_i)^{0.5}))$

（2）$T_2 = T_1/3$

これらの式において、T_1、T_2、H_i、G_i及びρ_iはそれぞれ次の値をあらわすものとする。

T_1　表層地盤の一次卓越周期（単位　sec）

T_2　表層地盤の二次卓越周期（単位　sec）

H_i　地盤調査によって求められた地盤の各層の層厚（単位　m）

G_i　地震時における地盤の各層のせん断剛性を表すものとして、地震時に生じる地盤のせん断ひずみに応じて土質ごとに別表第一に示される低減係数を次の式によって計算したG_{0i}に乗じて得た数値

$$G_{0i} = \rho_i V_{Si}^2$$

この式においてV_{Si}は、地盤調査によって求められた地盤の各層のせん断波速度（単位m/sec）を表すものとする。

ρ_i　地盤調査によって求められた地盤の各層の密度（単位t/m^3）

Translation

Calculation Formulas Used in the Dynamic Seismic Diagnosis Method

1. **Earthquake Intensity Used for the Assumed Seismic Conditions**

The following formula is used to estimate the earthquake intensity under assumed seismic conditions.

$$\log A = b_i - \log(X + c_i) - kX$$

Where: A: Maximum amplitude, X: Shortest distance from the fault plane (km), k: Viscous attenuation coefficient, b_i c_i: Regression coefficents determined for each earthquake

Next, the attenuation equation for maximum acceleration is obtained without considering the effect of surface ground conditions.

The coefficient k is fixed as: $k = 0.003$

The coefficients c_i and b_i are:

$$c_i = 0.000493 \times 10^{0.608Mw}$$

$$b_i = 0.495M_w + A_D + 0.479$$

where, M_w (Mw): Moment Magnitude, A_D:

$$\begin{cases} 0 & \text{for focal depth less than 30 km} \\ 0.186 & \text{for focal depth between 30 km and 60 km} \\ 0.554 & \text{for focal depth greater than or equal to 60 km} \end{cases}$$

Reference: Preliminary Study on Distance Attenuation Equations for Peak Ground Acceleration and Velocity Based on Japanese Earthquake Records Including the Hyogo-ken Nanbu Earthquake
Hirotoshi Tsukasa, Saburo Midorikawa, Masashi Matsuoka (1996)

2. **Natural Period and Amplification Ratio of Surface Ground**

According to Notification No.1457 of the Ministry of Construction, the first and second predominant periods of surface ground are calculated by the following equations.

(1) $T_1 = 4 \times (\Sigma H_i)^2 / (\Sigma((G_i/\rho_i) \times H_i)^{0.5})$

(2) $T_2 = T_1/3$

The symbols in these equations are defined as follows:

T_1: First predominant period of surface ground (sec)

T_2: Second predominant period of surface ground (sec)

H_i: Thickness of each soil layer determined by ground investigation (m)

G_i: Shear rigidity of each soil layer during earthquakes

$$G_{0i} = \rho_i V_{Si}^2$$

V_{Si}: Shear wave velocity of each soil layer obtained from ground investigation (m/sec)

ρ_i: Density of each soil layer obtained from ground investigation (t/m^3)

Readability Status

Readable (confirmed)

- Title is clearly readable.
- All equations are readable.
- Variable definitions are readable.
- Reference paper and authors are readable.

Partially Difficult to Read

- None

Not Readable / Unclear

- None

Original Document

２）表層地盤の一次卓越周期に対する増幅率 Gs_1 及び二次卓越周期に対する増幅率 Gs_2 は、それぞれ次に掲げる式によって計算するものとする。ただし、Gs_1 について、建築物の損傷限界時における値が１．５を下回る場合には１．５と、建築物の安全限界時における値が１．２を下回る場合には１．２と、それぞれするものとする。

（１）　$Gs_1 = 1 / (1.57 \times h + \alpha)$

（２）　$Gs_2 = 1 / (4.71 \times h + \alpha)$

これらの式において、α 及び h は、それぞれ次の数値を表すものとする。

α　次の式によって計算した波動インピーダンス比

$$\alpha = \Sigma \left((G_i / \rho_i)^{0.5} \times H_i \right) \times \Sigma (\rho_i \times H_i) / (\Sigma H_i)^2 \times (1 / (\rho_B \times V_B))$$

この式において、ρ_B 及び V_B は、それぞれ次の数値を表すものとする。

ρ_B　地盤調査によって求められた工学的基盤の密度(単位 t/m^3)

V_B　地盤調査によって求められた工学的基盤のせん断波速度（単位 m／sec）

h　地震時の表層地盤によるエネルギー吸収の程度を表すものとして次の式によって計算した数値（0.05 未満となる場合には、0.05 とする）

$$h = 0.8 \times (\Sigma (h_i \times W_i)) / \Sigma W_i$$

この式において、h_i 及び W_i は、それぞれ次の数値を表すものとする。

h_i　地震時における表層地盤の各層の減衰定数を表すものとして地震時に生ずる表層地盤のせん断ひずみ及び土質に応じて別表第二に示される数値

W_i　地震時における表層地盤の各層の最大弾性ひずみエネルギーを表すものとして次の式によって計算した数値

$$W_i = (G_i / (2 \times H_i)) \times (u_i - u_{i-1})$$

この式において、u_i は、地震時における地盤の各層における最上部の工学的基盤からの相対変位（単位 m）をあらわすものとする。

３．建物の状態

構造物における起振機による変位共振曲線と地表面の振動による構造物の変位共振曲線では、卓越周期における振幅の近似が認められる（耐震工学入門　平井一男・水田洋司　森北出版刊）。

ここから、建物の１次卓越周波数における応答倍率Ｒは、起振力と建物の応答した力との比をとることにより求めることができる。

$$R = (W_h A_h) / (W_e A_e)$$

W_h ：建物荷重

A_h ：建物２階床で検出された加速度

W_e ：起振機の荷重

A_e ：起振機の加速度

従って、上記で得た応答倍率Ｒを用いて、建物１階の壁が 1/120rad 変位するときの地表面の加速度 A_g は、以下の通りである。

$$A_g = (2\pi f)^2 \cdot d / R$$

f ：建物の１次卓越周波数

d ：建物が 1/120rad 変位するときの１階壁の変位量(cm)

Translation

2) Amplification Ratios Cormesponding to the First and Second Predominant Periods of Surface Ground

The amplification ratio for the first predominant period of the surface ground, Gs_1, and that for the second predominant period, Gs_1, are calculated using the following equations.

However, if the value of Gs_1 at the building damage limit state is less than 1.5, it is taken as 1.5; if the value at the building safety limit state is less than 1.2, it is taken as 1.2.

（１）　$Gs_1 = 1 / (1.57 \times h + \alpha)$

（２）　$Gs_2 = 1 / (4.71 \times h + \alpha)$

Where α and h are defined as:

α : Wave Impedance Ratio

$$\alpha = \Sigma ((G_i / \rho_i)^{0.5} \times H_i) \times \Sigma (\rho_i \times H_i) / (\Sigma H_i)^2 \times (1 / (\rho_B \times V_B))$$

where, ρ_B : Density of engineering bedrock obtained from ground investigation (t/m^3)

V_B : Shear wave velocity of engineering bedrock obtained from ground investigation (m/sec)

h : Damping Coefficient

represents the degree of energy absorption of the surface ground during earthquakes and is calculated as: (If the calculated value is less than 0.05, then 0.05 is used.)

$$h = 0.8 \times (\Sigma (h_i \times W_i)) / \Sigma W_i$$

where, h_i : Damping coefficient of each surface soil layer during earthquakes.

Wp, W_i : Maximum elastic strain energy of each surface soil layer during earthquakes.

$$W_L = (G_i / (2 \times H_i)) \times (u_i - u_{i-1})$$

where, u_i is the relative displacement of the upper boundary of each soil layer with respect to the engineering bedrock during earthquakes (m).

3. Condition of the Building

The displacement resonance curve obtained from excitation tests and the displacement resonance curve caused by ground vibration show similar amplitudes around the predominant period.

(*Introduction to Earthquake Engineering by Kazuo Hirai and Hiroshi Nagata, Morikita Publishing)

Therefore, the response amplification ratio R at the first predominant frequency of the building is obtained from the ratio between the excitation force and the building response.

$$R = (W_h A_h) / (W_e A_e)$$

W_h : Weight of the building

A_h : Acceleration detected on the second floor of the building

W_e : Weight of the exciter

A_e : Acceleration of the exciter

Therefore, using the response amplification ratio R, the ground surface acceleration A_g when the first-floor wall of the building deforms by 1/120 rad is given by:

$$A_g = (2\pi f)^2 \cdot d / R$$

where, f : First predominant frequency of the building

d : Displacement of the first-floor wall when the building deforms by 1/120 rad (cm)

Readability Status

Readable (confirmed)

- All explanatory text is clearly readable.
- All equations are readable.
- Variable definitions are readable.
- Reference literature is readable.

Partially Difficult to Read

- None

Not Readable / Unclear

- None